Task Hub system guide

How the system works and how to use it

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Task Hub is a Windows desktop application for preparing reports and repeating email tasks. A user connects an email account, saves a routine, and chooses whether Task Hub should prepare a draft or send automatically at the scheduled time. ExcelFlow is included to turn DISC and FTG workbooks into reports that can be attached to those routines.

This guide explains the current system in plain language. It is intended for users, colleagues testing the app, and anyone maintaining the project. It describes the local desktop system; it does not assume an always-running cloud service.

What users can do

Area	What it does
Connect	Connect Gmail, or an incoming IMAP/POP3 account with optional outgoing SMTP settings.
Email routines	Save recipients, optional CC, subject, message template, attachments, dates and repeat settings.
ExcelFlow	Read raw workbooks from email or a selected file, validate them, and generate DISC/FTG reports.
History and updates	Check execution results and install signed updates from published GitHub releases.

The overall flow

Connect an account → prepare report files if needed → save an email routine → wait for the schedule → create a draft or send → check email history.

What is currently unavailable

Microsoft 365 and Outlook Classic connections are temporarily unavailable. Previously saved Microsoft routines are preserved and paused. Customer Ageing and direct D365 import are not available. The system currently runs on a Windows computer; mobile control and an always-on cloud scheduler are not included.

How an email routine works

1 Connect the sending account

Choose Gmail in Connect and complete Google sign-in. For another provider, enter its IMAP or POP3 settings. Add SMTP settings if that account will send email. IMAP and POP3 receive email; SMTP sends it. A separate Task Hub user account is not required.

2 Save the email once

Create an email routine and choose the connected sender. Enter the recipient, optional CC, subject and message. Add any attachments, then choose the first date, time and repetition: once, daily, weekly or monthly. Saving the routine avoids retyping the same email each time.

3 Let the template fill in the period

The subject and message support {{month}}, {{year}}, {{date}}, {{previous_month}} and {{previous_year}}. These values use the routine occurrence date. For example, a routine due on 1 December 2026 with subject “{{previous_month}} {{previous_year}} report” becomes “November 2026 report”. Report numbers and explanations still need to come from the user; Task Hub does not invent report content.

4 Handle a change of recipient

Use the recipient-change settings to specify a replacement recipient and its start date. An end date makes the change temporary; leaving it open-ended keeps it in effect. Review these settings when responsibility changes. Edit the saved routine to make a permanent change to its default recipient.

5 Choose draft or automatic sending

Draft mode lets the user prepare the due email and then review or send it in their mailbox. Automatic sending uses the saved content and recipient when the routine is due, without another approval. Enabling automatic sending therefore authorizes future scheduled sends with those saved details.

6 Check the result

Task Hub validates the request, prepares attachments, contacts the email provider and records the outcome in Email history. Failed or uncertain sends are not automatically retried. If the result is uncertain, inspect Drafts and Sent before trying again to avoid duplicate emails.

Keep the computer awake and online. Closing the window keeps Task Hub running in the tray. Choosing Quit Task Hub, shutting down or sleeping the computer stops execution. Missed automatic occurrences may need rescheduling. Gmail can optionally add a routine reminder to Google Calendar after calendar access is enabled.

How ExcelFlow prepares reports

ExcelFlow processes DISC and FTG workbooks using the existing Harrison ExcelFlow report engine. Its calculations remain in one bundled engine rather than being rewritten inside the interface. No separate ExcelFlow application folder needs to be selected.

1 Open the report workspace

Choose ExcelFlow in the sidebar, then DISC & FTG Reports. The sidebar collapses to leave more room for the report screen. Use the top-bar button to expand it again. Workbooks are on the left, Report details on the right, and Output location below.

2 Choose the raw workbooks

Browse for a DISC workbook, an FTG workbook, or both. Alternatively, choose Read from email, select Gmail or IMAP/POP3, and browse recent emails. Assign the required workbook attachments as DISC or FTG. Gmail needs the optional attachment-reading permission. Only selected workbook attachments are downloaded; browsing does not delete messages or mark them read.

3 Set the reporting period and output

Choose which workbooks to process and whether to create Normal, Special or both report versions. Automatic month detection expects filenames such as ALL_DISC_2610.xlsx or ALL_FTG_2610.xlsx for October 2026. Other names, including oriALL_ prefixes, need a correctly named copy or a manually selected period in Settings.

Choose where the output should be saved. Settings also provides decimal precision, date recovery and the P160 multiplier. Manage IG names lets users maintain the agency-name mappings used for later report runs.

4 Validate and generate

Use Validate & preview to check dates without generating reports. View run details shows the date preview, run log and issues. Generate reports creates a new output folder. Correct any input errors before proceeding. Generating reports does not send an email.

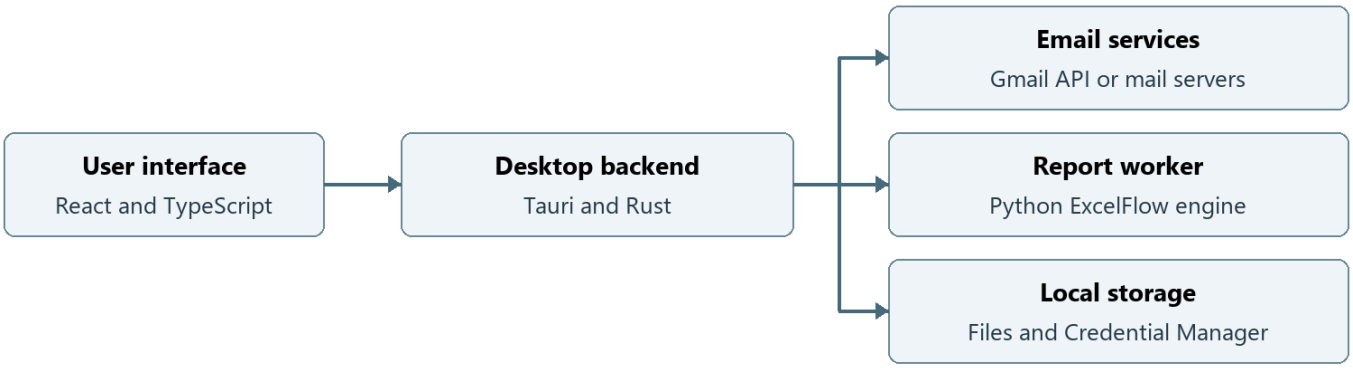
5 Add the results to an email routine

Choose Open results, select the completed run and the required reports, then choose a new or existing email routine. Add the selected reports and save the routine. Raw inputs can be up to 100 MB each. Outgoing attachments are limited to 10 files and 18 MB in total.

Important: Task Hub saves attachment copies. A later report run does not replace those copies automatically. Before the next month’s send, generate the new reports and replace the routine’s attachments. New Report Rules controls have been removed; any older rules remain active until explicitly removed from their saved routine.

System architecture and technology

The interface collects user choices. The Rust backend validates those choices, manages schedules and calls the appropriate email or report service. A bundled Python worker handles report generation and standard mail-server operations. No browser clicking is needed for these background email connections.



The technology stack

Layer	Technology	Purpose
Interface	React 19, TypeScript, HTML and CSS	Forms, routines, report controls and readable screens.
Desktop app	Tauri 2 and Rust	Windows app shell, validation, schedules, files and secure native commands.
Report processing	Python, openpyxl and the Harrison ExcelFlow engine	Read Excel workbooks, validate input and generate formatted reports.
Email	Gmail API with OAuth; TLS IMAP/POP3 and SMTP	Connect accounts and read attachments, create drafts or send emails.
Local data	JSON files, attachment copies and Windows Credential Manager	Store routines, history and files locally; protect connection credentials.
Build and delivery	Vite, PyInstaller, NSIS and the Tauri updater	Build the interface, bundle Python, create the Windows installer and verify updates.
Quality checks	Vitest, Playwright, Rust and Python tests	Check routine behavior, screens, native logic and report processing.

The app is modular: screens live in src/components, shared rules in src/domain, native-call helpers in src/services, Rust modules in src-tauri/src, and the bundled worker in integrations/runtime. This separation keeps interface changes away from report calculations and email transport logic.

Storage operation and maintenance

Where information is kept

Routines, schedules and history are stored as JSON in Task Hub's local application-data folder. Generated files are kept in its outputs folder or a user-selected output folder. Selected attachment copies and imported workbook inputs are also stored locally. Ordinary Gmail/IMAP messages remain with the email provider; Task Hub is not a full mailbox backup.

Gmail tokens and mail-server credentials use Windows Credential Manager. Email content, reports and routine data still exist as local files, so access to the Windows account and any backups should be controlled. No central multi-user database or cloud scheduler is part of the current design.

What happens when an email is prepared

Gmail creates a draft or submits the message through its API. IMAP can upload a draft to the configured Drafts folder. POP3 cannot upload drafts, so Task Hub saves a local .eml draft. SMTP can submit a message but does not automatically save a Sent Items copy. A successful submission is not a guarantee that the recipient has received or read it.

Keeping the app reliable

- Before each reporting period, check recipients, temporary overrides, template content and the actual attachment copies.
- Keep the computer awake, Task Hub running and the account connected. Use Email history to investigate failures before retrying.
- Back up local routine data and required reports. Keep the private update-signing key outside the source repository.
- After changing report logic or delivery behavior, run the relevant automated checks and test with a non-production routine.

How software updates reach users

Developers change and test the source, increase the version, build a signed installer and publish its installer, signature and update manifest in GitHub Releases. The installed app checks that published feed and offers verified newer versions. Local source edits do not change an already installed app or an earlier release.

For source development, Python 3.12 is needed to create the bundled report engine. The setup script creates an isolated environment, installs pinned dependencies and reuses unchanged builds. People using the installer do not need to install Python. Fully quit the tray app before launching a development copy because Task Hub allows only one running instance.

Possible future extensions

A cloud scheduler could support operation when the work computer is off. Mobile access, shared team storage, renewed Microsoft support and automatic monthly attachment replacement would each require additional development and testing. They should not be assumed available in v0.7.1.

Document basis: the local Task Hub v0.7.1 source, release notes and configuration reviewed on 10 November 2026. Future releases may change the behavior described here.